



Final Exam, Theory of Computation 2025

1 (15 pts) **Quick-fire round.** Consider the following statements.

1. If $A \leq_p B$, then $A \leq_m B$.
2. If $L \leq_p \text{SAT}$, then $L \leq_p \text{CLIQUE}$.
3. If $L \subseteq \text{SAT}$ then $L \in \mathbf{NP}$.
4. If $\text{SAT} \notin \mathbf{coNP}$, then $\mathbf{NP} \neq \mathbf{coNP}$.
5. If $\text{GRAPH-ISOMORPHISM} \in \mathbf{P}$, then $\mathbf{P} = \mathbf{NP}$.
6. If L is recognisable, then $L \leq_m \text{HALT}$.
7. If L is decidable, then L^* is decidable.
8. If $A \cup B$ is recognisable, then either A or B is recognisable.
9. Every boolean circuit can be equivalently written as a CNF formula.
10. Language $\{0^n 1^m : n \neq m\}$ is regular.
11. Language $\{x \in \{0, 1\}^* : x \text{ has the same number of occurrences of } 01 \text{ and } 10\}$ is regular.
12. Language $\{\langle \varphi \rangle : \varphi \text{ is a CNF that has } \geq 2025 \text{ satisfying assignments}\}$ is $\mathbf{NP}$ -complete.
13. Language $\{\langle G, k \rangle : G \text{ is a graph with no vertex cover of size } k\}$ is $\mathbf{NP}$ -complete.
14. Language $\{\langle M \rangle : M \text{ is a TM and } L(M) \text{ is regular}\}$ is decidable.
15. Language $\{\langle n, m \rangle : \text{BUSYBEAVER}(n) \geq m\}$ is recognisable.

For each statement, write down one of

- **T** if the statement is known to be true.
- **F** if the statement is false *or not known to be true*. E.g., both $\mathbf{P} = \mathbf{NP}$ and $\mathbf{P} \neq \mathbf{NP}$ should be marked **F**.
- or leave your answer empty.

A correct **T/F** answer is worth +1 point, an incorrect answer is worth −1 point, and an empty answer is worth 0 points.

Solution:

1.	2.	3.	4.	5.	6.	7.	8.	9.	10.	11.	12.	13.	14.	15.
T	T	F	T	F	T	T	F	T	F	T	T	F	F	T

2 (15 pts) Regular languages I.

2a Write down the statement of the Pumping Lemma. (*Only the statement, not its proof!*)

2b Consider strings over the alphabet $\Sigma = \{[,]\}$ and the language

$$L = \{x \in \Sigma^* : x \text{ is well-nested}\}.$$

For example, $[] [] \in L$, $[[] [[]]] \in L$, but $[]] \notin L$. Show that L is not regular.

Solution:

2a If A is a regular language, then there is a number p (the pumping length) such that, for every string s in A of length at least p , there exists a division of s into three pieces, $s = xyz$ s.t.

- for each $i \geq 0$, $xy^iz \in A$
- $|y| \geq 1$, and
- $|xy| \leq p$.

Grading scheme:

- Correct statement - 5 pts
- If the statement is not correct and the issue is minor - typically 4.5 pts.
- Otherwise - varies from case to case.

2b Suppose that L is a regular language. Then, let p be its pumping length. Consider the string $s = [^p]^p$, that is, the string which starts from p symbols $[$ and ends with p symbols $]$. Clearly, $s \in L$. We can apply the Pumping Lemma to s since $|s| \geq p$. By the Pumping Lemma, $|xy| \leq p$, from where both x and y are contained in the first p symbols $[$. Let k be the length of y : $y = [^k$. By the Pumping Lemma, $xz = [^{p-k}]^p \in L$, which is only possible if $k = 0$. This, however, contradicts the second property in the conclusion of the Pumping Lemma.

Grading scheme:

- Correct solution - 10 pts
- If the solution considers a particular partition of a string s into x, y and z – either 9.5pts or 7.5pts, depending on how much this limits the result
- Otherwise, varies from case to case and typically ≤ 6 pts.

3 (15 pts) NP-completeness I. A shy set in an undirected graph is a subset S of nodes such that each node in S has at most one neighbour in S . Show that SHY is NP-complete:

$$\text{SHY} = \{ \langle G, k \rangle : G \text{ is a graph that has a shy set of size } k \}.$$

Solution: The solution is almost identical to the one from Exercise X.3 (well-separated matchings).

NP-membership: given a set of vertices, one can check if it is a shy set in polytime (for each vertex, check that it has at most one neighbour in the set).

NP-hardness: given an instance of independent set problem $(G = (V, E), k)$ we generate an instance of SHY $(G', n + k)$ where $n = |V|$ in the following way. Graph G' is a copy of the graph G with extra edges (v, v') for every vertex v , where v' is a new vertex. Then if there is an independent set $S \subseteq G$ of size k , a set $S' = S \cup \{v' \mid v \in S\}$ is a shy set in G' of size $n + k$.

In the other direction, consider a shy set $S' \subseteq G'$ of size at least $n + k$. It consists of edges and individual vertices. Note that for every edge $e = (v, u)$, $e \in G$ we can replace u with v' (by the properties of the shy set, v' was not present there before), preserving the properties of the shy set and its size. In this way, we can make sure $E(S') \cap E = \emptyset$. Then $S = S' \cap G$ is an independent set in G of size at least k .

Grading scheme: 5 points for correct proof of NP-membership, 10 points for correct proof of NP-hardness.

4 (20 pts) NP-completeness II. Prove that SAT remains NP-complete under each of the following conditions on the input:

- 4a** No clause contains both a positive literal and a negative literal.
(Hint: Use a 2-CNF to express that two variables are assigned opposite bits.)
- 4b** No variable occurs (positively or negatively) in more than three clauses.

Solution: For both subproblem, it suffices to give a polynomial reduction from general SAT to the decision problem on the restricted formulas.

4a Let φ be any SAT formula over the variables $x_1, x_2, \dots, x_k$. Let $y_1, y_2, \dots, y_k$ be k additional variables and consider the formula

$$\psi = (x_1 \vee y_1) \wedge (\overline{x_1} \vee \overline{y_1}) \wedge (x_2 \vee y_2) \wedge (\overline{x_2} \vee \overline{y_2}) \wedge \dots \wedge (x_k \vee y_k) \wedge (\overline{x_k} \vee \overline{y_k}).$$

Note that ψ is true if and only if, for every $1 \leq i \leq k$, we have that y_i is the negation of x_i . Also note that no clause in ψ mixes positive and negative literals. Assuming that ψ holds, we can easily construct a formula φ' that is equivalent φ and where no clause contains both positive and negative literals (for instance by replacing every $\overline{x_i}$ with y_i and every $\overline{y_i}$ with x_i). Thus,

$$\varphi' \wedge \psi \text{ is satisfiable} \iff \varphi \text{ is satisfiable.}$$

Since we can construct $\varphi' \wedge \psi$ from φ in polynomial time, this gives the desired reduction.

4b Let φ be any SAT formula over the variables $x_1, x_2, \dots, x_k$. For every $1 \leq i \leq k$, let a_i denote the number of times x_i occurs in φ , where we can assume that $a_i \geq 1$. Now for each i , let $y_1^i, y_2^i, \dots, y_{a_i}^i$ be a_i many additional variables and consider the formula

$$\psi_i = (y_1^i \vee \overline{y_2^i}) \wedge (y_2^i \vee \overline{y_3^i}) \wedge \dots \wedge (y_{a_i-1}^i \vee \overline{y_{a_i}^i}) \wedge (y_{a_i}^i \vee \overline{y_1^i}).$$

Note that ψ_i is true if and only if all the variables $y_1^i, y_2^i, \dots, y_{a_i}^i$ are set to the same truth-value. Now, let φ' be the formula obtained from φ , but with the a_i instances of x_i replaced by the variables $y_1^i, y_2^i, \dots, y_{a_i}^i$. It follows that

$$\varphi' \wedge \psi_1 \wedge \psi_2 \wedge \dots \wedge \psi_k \text{ is satisfiable} \iff \varphi \text{ is satisfiable.}$$

Hence, since the right-hand side does not contain any variable more than thrice and is constructible in polynomial time from φ , this gives the desired reduction.

Grading scheme: No arguments are needed that the problems remain inside NP.

- Part a: 10 pts
 - Idea of introducing the y_i : 3 pts
- Part b: 10 pts.
 - Idea of introducing the y_j^i : 3 pts
 - If a variable might occur up to 5 times: 8 pts

5 (20 pts) Undecidability. Consider the language

$$\text{EXP} = \{ \langle M \rangle : M \text{ is a TM that halts in time } \leq 2^{|x|} \text{ on any input } x \in \{0,1\}^* \}.$$

Here $|x|$ denotes the length of the string x . Classify EXP as one of (i) decidable, (ii) undecidable but recognisable, (iii) unrecognisable. Justify your answer with a proof.

Solution:

We will show that EXP is **unrecognizable** through a reduction argument. Recall that the halting language $\text{HALT} = \{ \langle M, w \rangle : M(w) \text{ halts} \}$ is undecidable. Hence, its complement $\overline{\text{HALT}} = \{ \langle M, w \rangle : M(w) \text{ does not halt} \}$ is unrecognizable. We will show that $\overline{\text{HALT}}$ is mapping-reducible to EXP (through a computable map), which will imply the unrecognizability of EXP.

We now describe the following reduction $\overline{\text{HALT}} \leq_f \text{EXP}$. Let f be the computable function which maps the pair $\langle M, w \rangle$ to a TM description $\langle T \rangle$ such that T halts in at most $2^{|x|}$ steps on input x if and only if $M(w)$ does not halt. We construct T as follows:

$T =$ "On input x : Maintain a counter on the tape that ensures the following conditions are met:

- T begins to simulate the computation of $M(w)$ and proceeds with the simulation for at most $2^{|x|}$ steps.
- If the simulation of $M(w)$ has not resulted in halting, then T halts.
- If the simulation of $M(w)$ results in halting within less than $2^{|x|}$ steps of T , then T enters an infinite loop.

Hence, if $M(w)$ halts, then for large enough inputs x , $T(x)$ would be able to simulate the full computation of $M(w)$, check that $M(w)$ halts, and enter into an infinite loop. Therefore, when $\langle M, w \rangle \notin \overline{\text{HALT}}$, then $T = f(\langle M, w \rangle) \notin \text{EXP}$. However, if $\langle M, w \rangle \in \overline{\text{HALT}}$, and $M(w)$ never halts, then T will always halt within $2^{|x|}$ steps for any input x . This completes the reduction.

Grading Scheme.

- Correct answer. (2 pts.)
- Idea for a reduction from an unrecognizable language. (4 pts.)
- Correct selection of an unrecognizable language. (4 pts.)
- Correct construction of a computable reduction map. (10 pts.)
 - Only 2.5 pts. if the constructed T completely misses the point.
 - Up to 5 pts. if the construction of T is “morally” correct but with big implementations holes.
 - Up to 7.5 pts. if there are small implementation holes in the description of T .

Alternative solution.

- Correct proof of undecidability but not unrecognizability. (10 pts.)
 - Idea for a reduction and correct selection of an undecidable language (5 pts.)
 - “Morally” correct reduction with big holes (2.5 pts.)
 - “Morally” correct reduction with small holes (5 pts.)
- Correct answer (2 pts.)
- Correct proof that $\overline{\text{EXP}}$ is recognizable. (5 pts.)
- Statement that $\overline{\text{EXP}}$ being recognizable and EXP being undecidable leads to EXP being unrecognizable. (3 pts.)

Correct solutions that mistakenly take $\text{EXP} = \{\langle M \rangle : \exists x \in \{0, 1\}^* \text{ s.t. } M \text{ halts in at most } 2^{|x|} \text{ steps}\}$ will be awarded full points as well. In this misreading of the problem, Exp is recognizable but undecidable. In such proofs, 5 pts. are awarded for recognizability and 15 pts. for undecidability. Partial points are given for partially correct solutions. Again, 2 pts. for correct answer, 5.5 pts. for reduction idea and correct selection of undecidable language, 7.5 pts. for correct reduction.

6 (15 pts) Regular languages II. Let $|x|_b$ denote the number of occurrences of bit $b \in \{0, 1\}$ in a binary string $x \in \{0, 1\}^*$. For every $n \geq 2$, define $L_{n,b} = \{x \in \{0, 1\}^* : |x|_b \bmod n = 0\}$.

6a Describe a DFA with n states that accepts $L_{n,b}$.

6b Show that any NFA accepting $L_{n,0} \cap L_{n,1}$ has at least n^2 states.

(Hint: If an NFA uses fewer than n^2 states, use a pigeonhole argument to find two (cleverly chosen) strings that lead to the same state.)

Solution:

6a Let the set of states be integers $\{0, 1, \dots, n-1\}$, let $q_{\text{start}} = 0$, $Q_{\text{accept}} = \{0\}$; $\delta(i, b) = (i+1) \bmod n$; $\delta(i, 1-b) = i$. We can show by induction on the length of x that $\delta(q_{\text{start}}, x) = |x|_b$. The base case is $x = \epsilon$, $\delta(q_{\text{start}}, \epsilon) = q_{\text{start}} = 0$. Then take $x = x'b$ where $x' \in \{0, 1\}^*$. Since $|x'| < |x|$ by the induction hypothesis we have $\delta(q_{\text{start}}, x') = |x'|_b \bmod n$ we have $\delta(q_{\text{start}}, x) = \delta(|x'|_b \bmod n, b) = (|x'|_b + 1) \bmod n = |x|_b \bmod n$ and for $x = x'a$ for $a \neq b$ we have $\delta(q_{\text{start}}, x) = \delta(|x'|_b \bmod n, a) = |x'|_b \bmod n = |x|_b \bmod n$.

6b Let $S := \{s_{ij} \mid i, j \in \{0, 1, \dots, n-1\}\}$ where $s_{ij} = 0^i 1^j$. Suppose that $\delta(q_{\text{start}}, s_{ij}) \ni q$. Then the only string $s \in S$ such that $\delta(q, s) \cap Q_{\text{accept}} \neq \emptyset$ is $s_{(n-i) \bmod n, (n-j) \bmod n}$. Then $\delta(q_{\text{start}}, s_{ij}) \cap \delta(q_{\text{start}}, s_{i', j'}) = \emptyset$ for $(i, j) \neq (i', j')$ as otherwise for any state q in the intersection there must be a path from q according to $s_{(n-i) \bmod n, (n-j) \bmod n}$ terminating in an accepting state. But then $s_{i', j' s_{(n-i) \bmod n, (n-j) \bmod n}}$ is accepted, which is a contradiction since either $i' + (n-i) \bmod n$ or $j' + (n-j) \bmod n$ is not 0 modulo n .

Thus, the sets $\delta(q_{\text{start}}, s)$ for $s \in S$ are non-empty and disjoint, so the total number of states must be at least n^2 .

Grading scheme

6a Correct construction with a reasonable explanation: 5 points.

6b Cautionary example: $L_{n,0} \cap L_{n,1} \cap (0^*1^*)$ has an automaton with $O(n)$ states (first check that the number of zeroes divides n , then as soon as the first 1-character is read check that the number of ones divides n). Hence, if your argument *only* deals with strings of form $0^a 1^b$, it is incorrect.

- An essay why you think this should be true: 2 points;
- Assuming that states have the form q_{ij} where i tracks $|x|_0$ and j tracks $|x|_1$ without proof and concluding that one can't miss any i, j : 3 points;
- Similar to the previous one, assuming the only way you can construct an automaton for the intersection is to multiply the ones for the intersected languages: 3 points;
- Applying the pumping lemma incorrectly 3 points;
- Applying an OP version of the pigeonhole principle "suppose that $f: [N] \rightarrow [N-1]$ then $f(1) = f(2)$ ". I.e. you say that some particular strings s and t must reach the same state: 4 points if there is no formal proof why this collision is problematic 5 points if there is;
- Finding a state such that two strings with distinct values of $(|x|_0, |x|_1)$ end up in, but failing to rigorously argue (need to have some heuristic arguments to qualify though) that it means that the NFA is invalid: 4 points;
- Proof that is only correct for a DFA 7 points;
- Correct proof: 10 points.